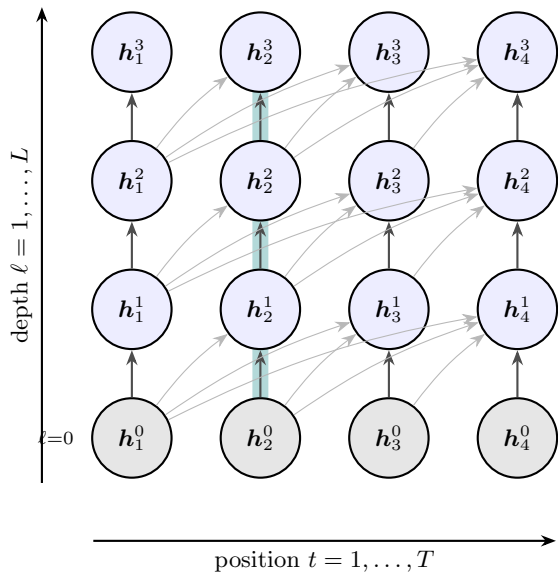
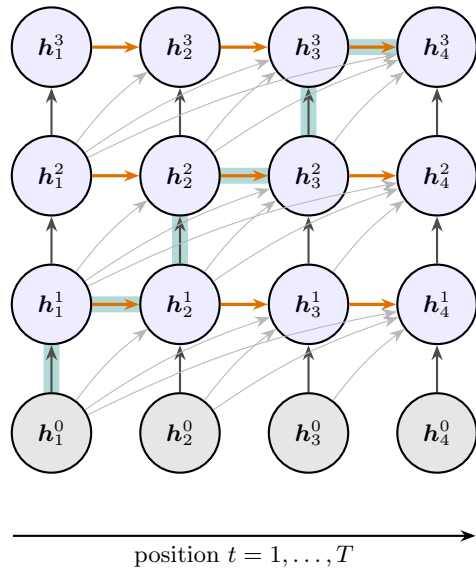


(a) attention only
opaque serial depth $\sim L$



(b) attention + horizontal recurrent step
opaque serial depth $\sim L + T$



→ attention (same position) → attention (earlier positions) → recurrent step (extra op) ■ longest opaque serial path